

45) Determine whether $\forall x(P(x) \rightarrow Q(x))$ and $\forall x P(x) \rightarrow \forall x Q(x)$ are logically equivalent.

Soln: Let $\forall x(P(x) \rightarrow Q(x))$ be False. By defn. of negation, $\neg(\forall x(P(x) \rightarrow Q(x)))$ is True. By De Morgan's Laws of quantifiers, $\exists x \neg(P(x) \rightarrow Q(x))$ is True. By conditional-disjunction equivalence, $\exists x \neg(\neg P(x) \vee Q(x))$ is True. By second De Morgan's Law, $\exists x (\neg(\neg P(x)) \wedge \neg Q(x))$ is True. By Double negation law, $\exists x (P(x) \wedge \neg Q(x))$ is True. $\therefore$ By defn, there exists an element in the domain s.t. $P(a) \wedge \neg Q(a)$ is True. By defn. of (And), $P(a)$ is True and $\neg Q(a)$ is True. By defn. of negation, $\neg Q(a)$ is True. By defn. of universal quantifier, $\forall x Q(x)$ is False. $\therefore a$ is in the domain. By defn of universal quantifier, $\forall x P(x)$ is True. We cannot conclude anything about $\forall x P(x) \rightarrow \forall x Q(x)$ from being True. It can be False, then by defn. of conditional, $\forall x P(x) \rightarrow \forall x Q(x)$ will be True. It can be True, if for all elements in the domain, $P(x)$ is individually True. Then by defn. of conditional, $\forall x P(x) \rightarrow \forall x Q(x)$ will be False.

$\therefore$ They are not logically equivalent $\square$

46) Determine whether $\forall x(P(x) \leftrightarrow Q(x))$ and $\forall x P(x) \leftrightarrow \forall x Q(x)$ are logically equivalent.

Soln: Let $\forall x(P(x) \leftrightarrow Q(x))$ be False. By defn. of negation, $\neg \forall x(P(x) \leftrightarrow Q(x))$ is True. By De Morgan's Laws of quantifiers, $\exists x \neg(P(x) \leftrightarrow Q(x))$ is True. From Ex-1.9, Prob 20, $\exists x \neg((P(x) \wedge Q(x)) \vee (\neg P(x) \wedge \neg Q(x)))$ is True. By 2nd De Morgan's Law, $\exists x (\neg(P(x) \wedge Q(x)) \wedge \neg(\neg P(x) \wedge \neg Q(x)))$ is True. By 1st De Morgan's Law, $\exists x (\neg(P(x) \wedge Q(x)) \wedge (\neg(\neg P(x)) \vee \neg(\neg Q(x))))$ is True. By double negation law, $\exists x (\neg(P(x) \wedge Q(x)) \wedge (P(x) \vee Q(x)))$ is True.